

# Erdős Problem #420

EP-420

L1 - TRACTABLE

OPEN

Number Theory

## ORIGINAL PROBLEM STATEMENT

If  $\tau(n)$  counts the number of divisors of  $n$  then let  $F(f, n) = \frac{\tau((n+[f(n)]))!}{\tau(n!)}$ . Is it true that  $\lim_{n \rightarrow \infty} F((\log n)^C, n) = \infty$  for large  $C$ ?

Is it true that  $F(\log n, n)$  is everywhere dense in  $(1, \infty)$ ?

More generally, if  $f(n) \leq \log n$  is a monotonic function such that  $f(n) \rightarrow \infty$  as  $n \rightarrow \infty$ , then is  $F(f, n)$  everywhere dense?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>